Activity 8

1. **Fourier Series:** Consider a function that is periodic with period τ. This means after some time τ, the function repeats itself, or in other words:

$$f(t+\tau) = f(t)$$

Fourier's theorem states that you can write any τ-periodic function as a sum of sines and cosines:

$$f(t) = \sum_{n=0}^{\infty} \left[a_n \cos(n\omega t) + b_n \sin(n\omega t)\right] ,$$

where $\omega = 2\pi/\tau$. The trick is to find the right a_n, b_n, which are determined by $f(t)$. You can find these coefficients using:

$$a_0 = \frac{1}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\,, \qquad [n=0]$$

$$a_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\cos(n\omega t)dt\,, \qquad n>0$$

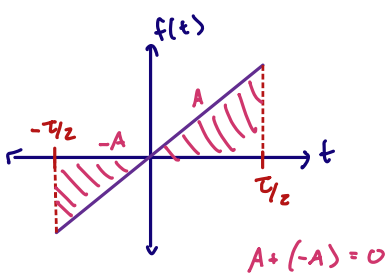
$$b_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\sin(n\omega t)dt\,, \qquad n>0 \qquad (n=0 \text{ just vanishes due to } \sin(n\omega t))$$

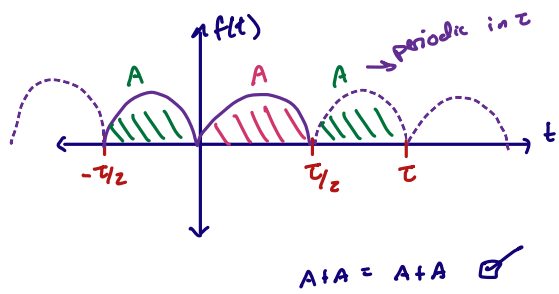
(a) Draw a picture that demonstrates that if $f(t)$ is odd,

$$\int_{-\tau/2}^{\tau/2} f(t)^{(\text{odd})}dt = 0$$

and a picture that demonstrates that if $f(t)$ is even,

$$\int_{-\tau/2}^{\tau/2} f(t)^{(\text{even})}dt = \int_0^{\tau} f(t)dt$$





(b) For each of the following, say whether the integrand is odd, even, or neither. If it's odd or even, simplify/rewrite as per part (1a).

- $2/\tau \int_{-\tau/2}^{\tau/2} f(t)^{(\text{odd})} \cos(n\omega t)dt$ $= 0$ odd
- $2/\tau \int_{-\tau/2}^{\tau/2} f(t)^{(\text{odd})} \sin(n\omega t)dt$ $= 2/\tau \int_0^{\tau} f(t)^{\text{odd}} \sin(n\omega t)dt$ even
- $2/\tau \int_{-\tau/2}^{\tau/2} f(t)^{(\text{even})} \cos(n\omega t)dt$ $= 2/\tau \int_0^{\tau} f(t)^{\text{odd}} \sin(n\omega t)dt$ even
- $2/\tau \int_{-\tau/2}^{\tau/2} f(t)^{(\text{even})} \sin(n\omega t)dt$ $= 0$ odd

(c) Expand the function shown in Fig. 1 as a Fourier series:

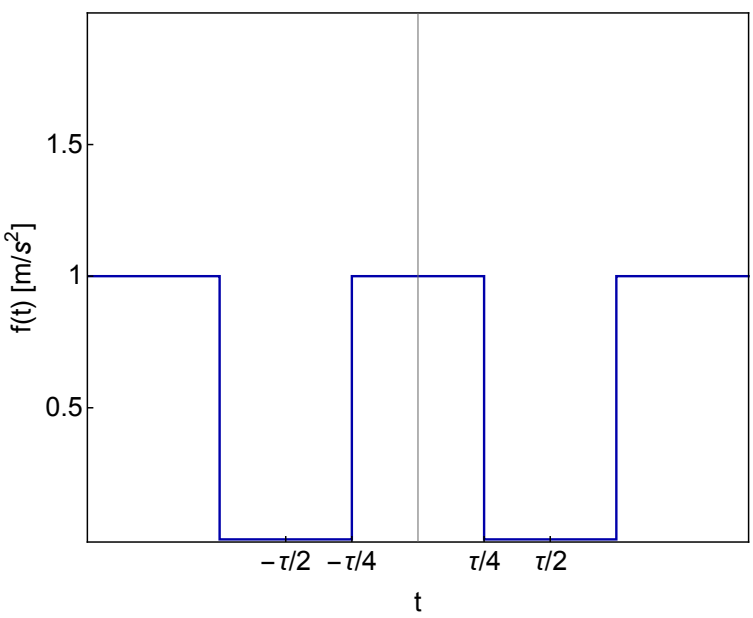


Figure 1

$$a_0 = \frac{1}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\,dt = \frac{1}{\tau}\int_{-\tau/4}^{\tau/4} dt = \frac{1}{\tau}(t)\Big|_{-\tau/4}^{\tau/4} = \frac{1}{2}$$

$b_n = 0$, odd

$$a_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\cos(n\omega t)\,dt = \frac{2}{\tau}\int_{-\tau/4}^{\tau/4}\cos(n\omega t)\,dt = \frac{2}{\tau}\frac{1}{n\omega}\sin(n\omega t)\Big|_{-\tau/4}^{\tau/4} = \frac{2}{\tau}\frac{1}{n\omega}\,2\sin\left(\frac{n\omega\tau}{4}\right)$$

$\omega\tau = 2\pi$

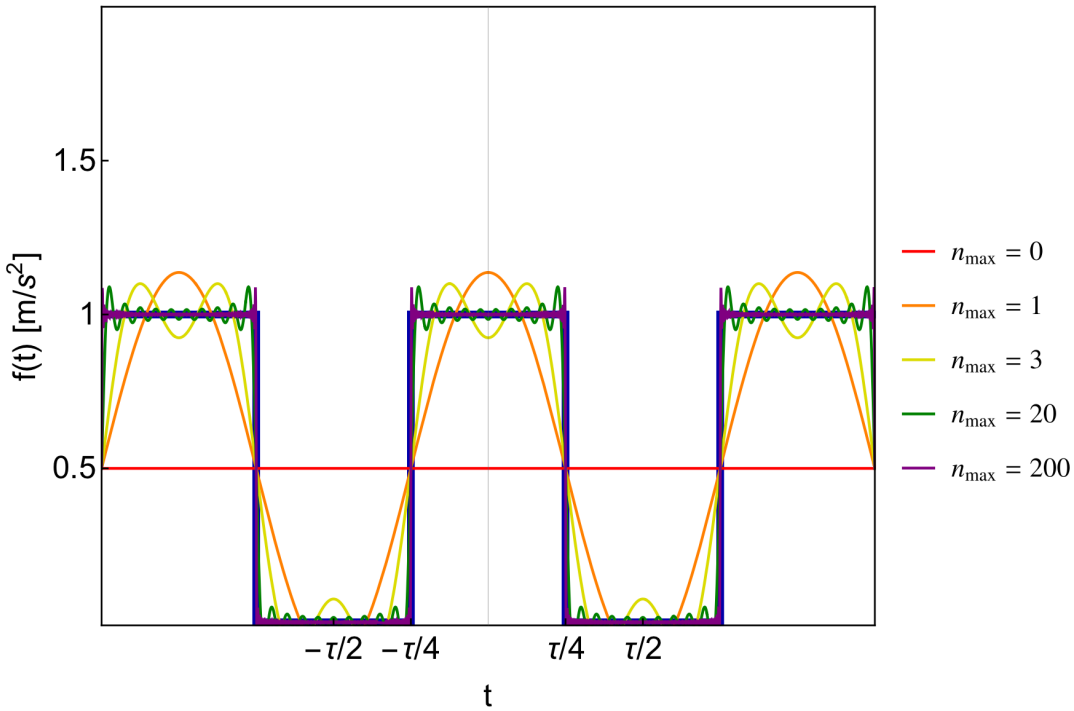
$$= \frac{4}{n\omega\tau}\sin\left(\frac{n\omega\tau}{4}\right) = \frac{2}{n\pi}\sin\left(\frac{n\pi}{2}\right) \Rightarrow \quad a_{2k} = 0, \quad a_{2k+1} = \frac{2}{(2k+1)\pi}(-1)^k$$

or

$$\Rightarrow f(t) = \frac{1}{2} + \sum_{k=0}^{\infty}\frac{2}{(2k+1)\pi}(-1)^k\cos\left[(2k+1)\omega t\right]$$

$$f(t) = \frac{1}{2} + \sum_{n=1}^{\infty}\frac{2}{n\pi}\sin\left(\frac{n\pi}{2}\right)\cos(n\omega t)$$

$$f(t) = \frac{1}{2} + \sum_{k=0}^{n_{max}} \frac{2}{(2k+1)\pi}(-1)^k \cos\left[(2k+1)\omega t\right], \text{ for various } n_{max}$$



You can see that as $n_{max} \to \infty$, $f(t)$ approaches the original function

Activity 6

(d) Expand the function shown in Fig. 2 as a Fourier series.

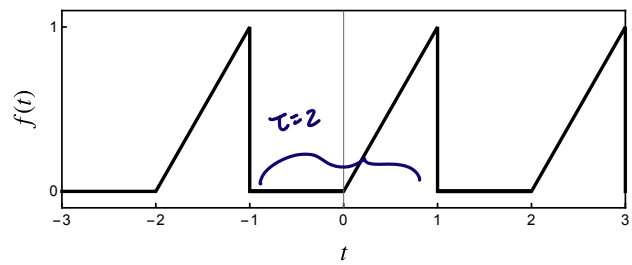


Figure 2

$$a_0 = \frac{1}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\,dt = \frac{1}{2}\int_0^1 t\,dt = \frac{1}{2}\,\frac{t^2}{2}\Big|_0^1 = \boxed{\frac{1}{4}}$$

$$\int u\cos u\,du = \cos u + u\sin u + C$$

$u = n\omega t$

$du = n\omega\,dt$

$$a_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\cos(n\omega t)\,dt = \frac{2}{2}\int_0^1 t\cos(n\omega t)\,dt$$

$$= \int_0^1 \frac{n\omega t}{n\omega}\cos(n\omega t)\,\frac{d(n\omega t)}{n\omega} = \frac{1}{(n\omega)^2}\int_{u_0}^{u_1} u\cos u\,du$$

$$= \frac{1}{n^2\omega^2}\Big[\cos u + u\sin u\Big]_{u_1}^{u_2} = \frac{1}{n^2\omega^2}\Big[\cos(n\omega t) + n\omega t\sin(n\omega t)\Big]_0^1 = \frac{1}{n^2\omega^2}\Big[\cos(n\omega) - 1 + n\omega\sin(n\omega) - 0\Big]$$

$$= \frac{1}{n^2\omega^2}\Big[\cos n\omega - 1 + n\omega\sin n\omega\Big] \qquad \omega = \frac{2\pi}{\tau} = \frac{2\pi}{2} = \pi$$

$$= \frac{1}{n^2\pi^2}\Big[\cos n\pi - 1 + n\pi\sin n\pi\Big]$$

$$= \frac{\cos n\pi - 1}{n^2\pi^2} = \frac{(-1)^n - 1}{n^2\pi^2} = \begin{cases} 0 & n = \text{even} \\ \dfrac{-2}{n^2\pi^2} & n = \text{odd} \end{cases}$$

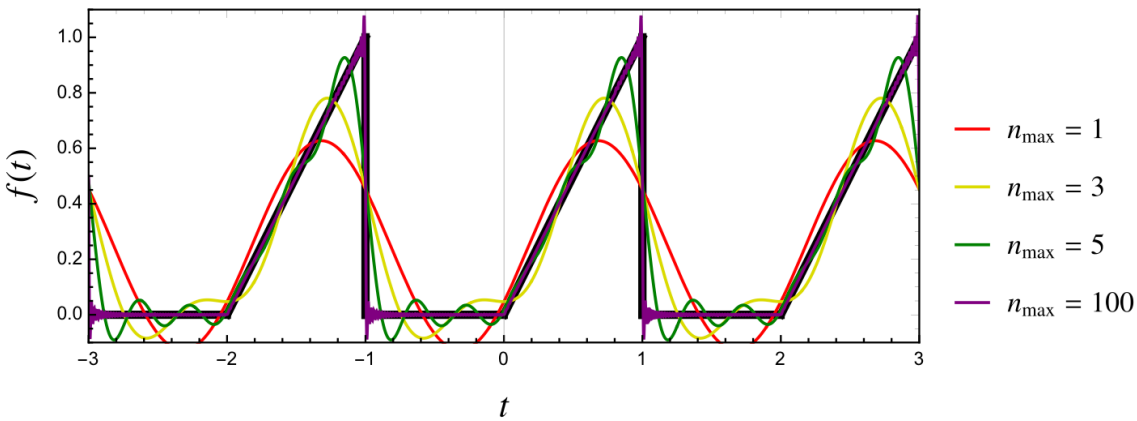
$$\int u\sin u\,du = -u\cos u + \sin u + C$$

$$b_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\sin(n\omega t)\,dt = \int_0^1 t\sin(n\omega t)\,dt = \frac{1}{n^2\omega^2}\Big[\sin(n\omega t) - n\omega t\cos(n\omega t)\Big]_0^1$$

$$= \frac{1}{n^2\omega^2}\Big[\sin(n\omega) - 0 - n\omega\cos(n\omega) + 0\Big] = \frac{1}{n^2\pi^2}\Big[\cancelto{0}{\sin(n\pi)} - n\pi\cos(n\pi)\Big] = \frac{(-1)^{n+1}}{n\pi}$$

$$f(t) = \frac{1}{4} + \sum_{n=1}^{\infty}\left[\frac{(-1)^{n+1}}{n\pi}\sin(n\pi t) + \frac{(-1)^n - 1}{n^2\pi^2}\cos(n\pi t)\right]$$

f(t) for various n_{max} :



You can see that as $n_{max} \to \infty$, f(t) approaches the original function